Optimal Scheduling for Charging and Discharging of Electric Vehicles

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***Abstract*—The vehicle electrification will have a significant impact on the power grid due to the increase in electricity consumption. It is important to perform intelligent scheduling for charging and discharging of electric vehicles (EVs). However, there are two major challenges in the scheduling problem. First, it is challenging to find the globally optimal scheduling solution which can minimize the total cost. Second, it is difficult to find a distributed scheduling scheme which can handle a large population and the random arrivals of the EVs. In this paper, we propose a globally optimal scheduling scheme and a locally optimal scheduling scheme for EV charging and discharging. We first formulate a global scheduling optimization problem, in which the charging powers are optimized to minimize the total cost of all EVs which perform charging and discharging during the day. The globally optimal solution provides the globally minimal total cost. However, the globally optimal scheduling scheme is impractical since it requires the information on the future base loads and the arrival times and the charging periods of the EVs that will arrive in the future time of the day. To develop a practical scheduling scheme, we then formulate a local scheduling optimization problem, which aims to minimize the total cost of the EVs in the current ongoing EV set in the local group. The locally optimal scheduling scheme is not only scalable to a large EV population but also resilient to the dynamic EV arrivals. Through simulations, we demonstrate that the locally optimal scheduling scheme can achieve a close performance compared to the globally optimal scheduling scheme.**

***Index Terms*—Charging and discharging, convex optimization, distributed solution, electric vehicle, optimal scheduling, smart grid, vehicle-to-grid (V2G).**

NOMENCLATURE

$\mathbf{N}$	Interval set.
$\mathbf{M}$	Set of electric vehicles (EVs),
$\mathbf{M}^{\mathrm{CHG}}$	Charging-only EV set.
$\mathbf{M}^{V2G}$	Vehicle-to-grid (V2G) EV set.
x_{mi}	Charging power of EV m in interval i.
T_m	Charging period of EV m.
τ	Length of an interval
E_m^{ini}	Initial energy of EV m.
E_m^{cap}	Battery capacity of EV m.
E_m^{fin}	Final energy of EV m.
$P^{\max}$	Maximum charging power.
γ_m	Final energy ratio of EV m.
$\mathbf{F}$	Charging-interval matrix.
z_i	Total load in interval i.
L_i^b	Real base load in interval i.
L_i^{bF}	Forecasted base load in interval i.
y_i	Charging load in interval i.
k_0	Intercept in the real-time pricing model.
k_1	Slope in the real-time pricing model.
C_i	Cost for EV charging in interval i.
$\mathbf{Q}^{(i)}$	Previous-interval set of interval i.
$\mathbf{B}$	Group set.
$\mathbf{H}_k^{(i)}$	Ongoing EV set at the beginning of interval i in group k.
$\mathbf{W}_k^{(i)}$	Sliding window at the beginning of interval i in group k.
$\mathbf{H}_k^{(i)\mathrm{CHG}}$	Charging-only EV set at the beginning of interval i in group k.
$\mathbf{H}_k^{(i)V2G}$	V2G EV set at the beginning of interval i in group k.
t_m^{arr}	Arrival time of EV m.
t_m^{dep}	Departure time of EV m.
t_m^{C-s}	Start time of the charging period of EV m.
t_m^{C-e}	End time of the charging period of EV m.

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I. INTRODUCTION

THE automotive industry is heavily investing in plug-in hybrid electric vehicles (PHEVs) and fully electric vehicles (EVs) mainly in order to reduce the CO_2 emissions and oil dependency of current automotive technology. The vehicle electrification will have significant impacts on the power grid due to the increase in electricity consumption.

The overall load profile of electric system will be changed due to the introduction of EV charging and discharging. The charging of a large population of EVs has a significant impact on the power grid. It have been estimated that the total charging load of the EVs in US can reach 18% of the U.S. summer peak at the EV penetration level of 30% [1]. On the other hand, an EV can also provide energy to the power grid by discharging

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the battery, which is known as vehicle-to-grid (V2G) [2]. An intelligent scheduling scheme can optimally schedule the EV charging patterns such that the load profile of the electric system can be effectively flattened. This will reduce potential capital costs and minimize operational costs. Intelligent scheduling for EV charging and discharging has become a vital step towards smart grid implementation [3], [4]. The essential principle in intelligent scheduling is to reshape the load profile by charging the EV battery from the grid at the time when the demand is low and discharging the EV battery to the grid when the demand is high. However, it is challenging to schedule the patterns of EV charging and discharging in an optimal way. First, it is difficult to find the globally optimal scheduling solution which can minimize the overall charging cost, especially in the presence of a large EV population. Second, the scheduling scheme is required to have the capacity to efficiently handle the random arrivals of the EVs.

In the recent literature, a number of scheduling schemes for EV charging and discharging have been proposed [5]–[8]. However, the scheduling schemes in [5], [6] only dealt with battery charging without V2G function. Though the existing work on V2G scheduling [7], [8] tried to optimize the charging and discharging powers to minimize the cost, their methods are essentially centralized algorithms, which may not be suitable for the EV charging and discharging systems with a large population and dynamic arrivals.

In this paper, we propose a globally optimal scheduling scheme and a locally optimal scheduling scheme for EV charging and discharging. Our contributions are summarized as follows.

- We formulate a global scheduling optimization problem, which aims to minimize the total cost for charging all EVs within the day. The optimization problem is a convex optimization problem, which can be solved efficiently. The globally optimal scheduling scheme determines the optimal charging powers for all EVs for all intervals by solving a single *global scheduling optimization problem*, thus obtaining the globally minimal total cost.
- We formulate a local scheduling optimization problem for the EVs in the local group. Based on the *local scheduling optimization problem*, we develop a locally optimal scheduling scheme, which is performed in an independent and distributed way. The locally optimal scheduling scheme is very appropriate for the EV charging and discharging systems with a large population and dynamic arrivals. The performance of the locally optimal scheduling scheme is lower than but very close to that of the globally optimal scheduling scheme.

The globally optimal scheduling scheme provides the globally minimal total cost. However, the globally optimal scheduling scheme is impractical since it requires the information on the future base loads and the arrival times and the charging periods of the EVs that will arrive in the future time of the day. Though the locally optimal scheduling scheme performs a little worse than the globally optimal scheduling scheme, it it is a practical scheme which can efficiently handle a large EV population and dynamic EV arrivals. Therefore, the locally optimal scheduling scheme is the final solution suggested in the paper. With the globally minimal total cost provided by the globally optimal scheduling scheme, we can find out the optimality gap between the two schemes.

The remainder of the paper is organized as follows. Section II discusses the related work. In Section III, we formulate and solve the global scheduling optimization problem. In Section IV, we formulate and solve the local scheduling optimization problem. The simulation results are presented in Section V, and the conclusions are drawn in Section VI.

II. Related Work

Depending on the direction of energy flow, existing work on EV charging scheduling can be classified into two classes: 1) scheduling for charging only, and 2) scheduling for both charging and discharging.

In charging-only scheduling, the scheduler tries to optimize the energy flow from the grid to the battery of the EV. In [5], Shrestha *et al.* optimized the EV battery charging during the low-cost off-peak period to minimize the charging cost in the context of Singapore. The paper in [9] examined the problem of optimizing the charge trajectory of a PHEV, defined as the time and the rate with which the PHEV obtains electricity from the power grid. In [1], a decentralized charging control algorithm was proposed to schedule charging for large populations of EVs. The paper in [10] optimized EV battery charging behavior to minimize charging costs, achieving satisfactory state-of-energy levels, and optimal power balancing. Mets *et al.* in [6] presented smart energy control strategies for charging residential PHEVs, aiming to minimize the peak load and flatten the overall load profile. The impact of different battery charging rates of EVs on the power quality of smart grid distribution systems was studied in [11]. In [12], Clement *et al.* proposed coordinated charging with stochastic programming, which was introduced to represent the error in the load forecasting.

In charging and discharging scheduling, the scheduler tries to optimize the bidirectional energy flows: from the grid to the EV battery and from the EV battery to the grid. Binary particle swarm methods were employed to optimize the V2G scheduling in a parking lot to maximize the profit [7], [8], [13]. Sortomme *et al.* proposed an unidirectional regulation at the aggregator, in which several smart charging algorithms were examined to set the point about which the rate of charge varies while performing regulation [14]. The paper in [16] developed an aggregator for V2G frequency regulation with the optimal control strategy, which aims to maximize the revenue. Jang *et al.* proposed a method for an analytic estimation of the probability distribution of the procured power capacity (PPC), based on which the optimal contract size was decided [17]. The paper in [18] presented a real-time model of a fleet of plug-in vehicles performing V2G power transactions. In [19], Singh *et al.* demonstrated that the coordinated charging and discharging of EVs can improve the voltage profile and reduce the power transmission loss. The paper in [15] discussed the vehicle to grid integration and described the vehicle-to-grid communication interface.

III. Global Scheduling Optimization

In this section, we formulate a global scheduling optimization for EV charging and discharging based on a real-time pricing

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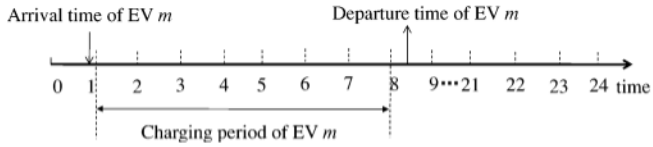


Fig. 1. Charging period of EV m.

model. The solution to the optimization problem provides a globally optimal scheduling scheme which minimizes the total cost.

A. System Models

We study the battery charging and discharging of EVs during a day, which is evenly divided into a set of intervals. The interval set is denoted by $\mathbf{N}$. The length of an interval is denoted by τ. We assume that the charging or discharging power in an interval is kept unchanged. In this paper, we divide the day into 24 intervals such that the interval length is given by $\tau = 1$ h.

The set of the EVs, which perform charging and discharging during the day, is denoted by $\mathbf{M}$. The EV set $\mathbf{M}$ consists of two sets: 1) the charging-only EV set $\mathbf{M}^{\mathrm{CHG}}$, which includes the EVs that only charge their battery and do not provide the battery energy to the grid, and 2) the V2G EV set $\mathbf{M}^{V2G}$, which includes the EVs that perform both battery charging and battery discharging. We have $\mathbf{M} = \mathbf{M}^{\mathrm{CHG}} + \mathbf{M}^{V2G}$. The charging or discharging power of EV m in interval i is denoted by x_{mi} $(\forall m \in \mathbf{M}, \forall i \in \mathbf{N})$. In order to unify the notation, we just call x_{mi} the charging power of EV m in interval i. If $x_{mi} > 0$, it means that EV m charges its battery in interval i. If $x_{mi} < 0$, it means that EV m discharges its battery in interval i. The EVs in the charging-only set $\mathbf{M}^{\mathrm{CHG}}$ always satisfy $x_{mi} \geq 0$ since they do not discharge their battery at any time. On the other hand, the EVs in the V2G set $\mathbf{M}^{V2G}$ may have a positive, zero, or negative charging power x_{mi} in interval i $(\forall i \in \mathbf{N})$ since they have bidirectional energy flows between the battery and the power grid.

The arrival time of EV m, denoted by t_m^{arr}, is the time when EV m is plugged into the charging station. The departure time of EV m, denoted by t_m^{dep}, is the time when EV m is plugged out of the charging station. The charging period of EV m, denoted by T_m, is the period in which EV m charges or discharges its battery. Since we divide the time into multiple intervals, we define the charging period T_m of EV m as the set of continuous intervals that fall between the arrival time t_m^{arr} and the departure time t_m^{dep} of EV m, as illustrated in Fig. 1. The initial energy of EV m, denoted by E_m^{ini}, is defined as the battery energy at the arrival time t_m^{arr}. The battery capacity of EV m is denoted by E_m^{cap}. The final energy of EV m, denoted by E_m^{fin}, is defined as the battery energy at the departure time t_m^{arr}. The final energy E_m^{fin} is no larger than the battery capacity E_m^{cap}. We define a *final energy ratio* of EV m as $\gamma_m = E_m^{\mathrm{fin}}/E_m^{\mathrm{cap}}$ where $0 \leq \gamma_m \leq 1$. The charging station can automatically detect the arrival time, the initial energy and the battery capacity of EV m when the EV is connected to the charging station. The departure time and the final energy ratio of EV m are provided to the charging station by the user of EV m before charging is started. The charging station can determine the charging period T_m of EV m from the parameters t_m^{arr} and t_m^{dep}. EV m performs charging and discharging activities during the charging period T_m. To represent the relationship between the charging/discharging activities and the intervals, we define a *charging-interval matrix* $\mathbf{F} \subset \{0,1\}^{|\mathbf{M}|\times|\mathbf{N}|}$ where $|\mathbf{M}|$ and $|\mathbf{N}|$ denote the number of elements in the set $\mathbf{M}$ and the set $\mathbf{N}$, respectively. The elements of $\mathbf{F}$ are defined as

$$f_{mi} = \begin{cases} 1, & \text{if interval } i \text{ falls within the charging period } T_m \\ & \text{of EV } m, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

In this paper, we consider the scheduling of EV charging and discharging in a small geographic area. In our real-time pricing model, we make two assumptions: 1) the losses between nodes are small and thus negligible, and 2) there is no congestion in transmission. The two assumptions allow us to neglect the spatial variation of the electricity prices. The electricity price at a time instant is the same regardless of the charging location. The optimizations of EV charging based on only temporal variation but not spatial variation of the price have be seen in [1], [6]. The electricity price is modeled as a linear function of the instant load [1], which is given as follows:

$$g(z_t) = k_0 + k_1 z_t, \tag{2}$$

where k_0 is the intercept and k_1 is the slope, which are both non-negative real number, and z_i is the total load at time t.

The total load in interval i consists of two parts: 1) the base load L_i^b, which represents the load of all electricity consumptions in interval i except EV charging, and 2) the charging load y_i, which represents the load of EV charging in interval i. We assume that the base load L_i^b remains constant in interval i. The charging load in interval i is given by $y_i = \sum_{m\in\mathbf{M}} x_{mi} f_{mi}$. If the load from the grid to the batteries of the EVs is greater than that from the batteries of the EVs to the grid in interval i, the charging load y_i is positive. Otherwise, it is negative. The total load in interval i is given by $z_i = L_i^b + y_i = L_i^b + \sum_{m\in\mathbf{M}} x_{mi} f_{mi}$. Since both the base load L_i^b and the charging power x_{mi} $(\forall m \in \mathbf{M}, \forall i \in \mathbf{N})$ remain constant in interval i, the total load z_i is constant in interval i.

In this paper, we define the charging cost in interval i, denoted as C_i, as the total amount of the money that the customers pay for charging and discharging of their EVs in interval i. Based on the pricing model, the charging cost in interval i $(\forall i \in \mathbf{N})$ is given by

$$\begin{aligned} C_i &= \int_{L_i^b}^{z_i} (k_0 + k_1 z_t) dz_t \\ &= \left(k_0 z_i + \frac{k_1}{2} z_i^2\right) - \left(k_0 L_i^b + \frac{k_1}{2} \left(L_i^b\right)^2\right). \end{aligned} \tag{3}$$

As shown in (3), the charging cost C_i can be positive or negative. If the charging load y_i, given by $y_i = z_i - L_i^b$, in interval i is positive, the charging cost C_i is positive. Otherwise, it is negative.

B. Problem Formulation and Solution

In order to find a globally optimal scheduling scheme for the EVs that perform charging and discharging during the day, we

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